Physics 4C Student's Name: Lotuseana Ng

Worksheet #3 Group: Tuesday/ Thursday

A monatomic ideal gas expands from $3.0\ l$ to $4.0\ l$ along a process defined by

$$P = \frac{a}{V^2} \quad \text{where } a = 10\ atm \times l^2$$

The initial temperature of the ideal gas is $300\ K$.

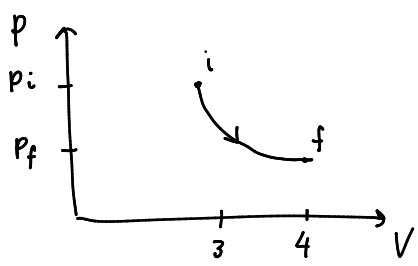
a. Express $1\ atm \times l^2$ in terms of SI units.

$1\ atm = 1.01 \times 10^5\ Pa \quad ; \quad l^2 = 10^3\ cm^3$

$$1\ atm \times l^2 = 1\ atm \cdot \frac{1.01 \times 10^5\ Pa}{1\ atm} \cdot l^2 \left(\frac{10^3\ cm^3 \cdot 1\ m^3}{10^6\ cm^3}\right)^2$$

$$= \frac{1.01 \times 10^5\ Pa}{(10^3\ m^3)^2} = \frac{1.01 \times 10^5}{10^6}\ Pa\ (m^3)^2 = \boxed{0.101\ Pa\ (m^3)^2}$$

b. Determine the initial and final pressure of the gas and sketch the process in a PV-diagram.



$P = \frac{a}{V^2} \qquad V_i = 3l \qquad V_f = 4l$

$$P_i = \frac{(10\ atm \cdot l^2)}{(3l)^2} = \frac{10}{9}\ atm$$

$$P_f = \frac{(10\ atm \cdot l^2)}{(4l)^2} = \frac{10}{16}\ atm$$

c. Calculate the number of molecules of the gas. How many moles does that correspond to?

$PV = NkT \qquad P_i = 1.12\ atm \cdot \frac{1.01 \times 10^5\ Pa}{1\ atm} = 112110\ Pa$

$N = \frac{P_i V_i}{k T_i} \qquad V_i = 3l = 0.003\ m^3$

$$N = \frac{(112110\ Pa)(3 \times 10^{-3}\ m^3)}{(1.38 \times 10^{-23})(300\ K)} = \boxed{8.12 \times 10^{22}\ \text{molecules}}$$

$$n = \frac{N}{N_A} = \frac{8.12 \times 10^{22}}{6.02 \times 10^{23}} = \boxed{0.135\ \text{moles}}$$

d. Determine the final temperature of the gas.

$P_f V_f = NkT_f$

$$T_f = \frac{(\frac{10}{16} \times 1.01 \times 105)(0.004\ m^3)}{(8.12 \times 10^{22})(1.38 \times 10^{-23})} = \boxed{225\ K}$$

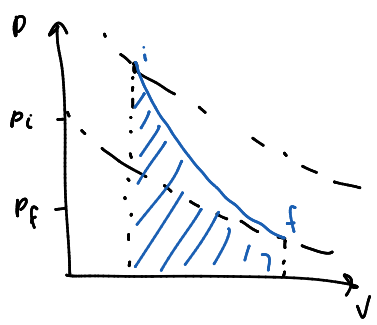
e. Calculate the change in the internal energy of the gas.

$\Delta E_{int} = \frac{3}{2} Nk\,\Delta T = \frac{3}{2} Nk(T_f - T_i) = \frac{3}{2} Nk\,(225-300)$

$\Delta E_{int} = \frac{3}{2}(8.12\times10^{22})(1.38\times10^{-23})(-75) = \boxed{-126.1\ J}$

f. Determine the net work done on / by the gas during this process.

$\Delta E_{int} = Q - W$



$W = \int_{V_i}^{V_f} P\,dV = \int_{V_i}^{V_f} \frac{a}{V^2}\,dV = a\int_{V_i}^{V_f} \frac{1}{V^2}\,dV$

$= \left[-a\frac{1}{V}\right]_{V_i}^{V_f} = -a\left[\frac{1}{V_f} - \frac{1}{V_i}\right]$

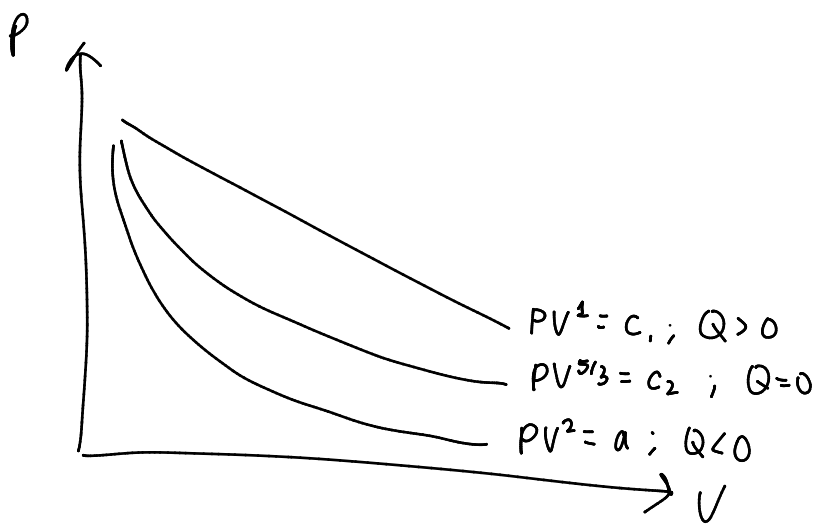
$= -a\left[\frac{1}{4} - \frac{1}{3}\right] = -a\left[-\frac{1}{12}\ell\right] = \frac{a}{12}\ell$

$= \frac{10\ atm\cdot \ell^2}{12\ \ell} = \frac{5}{6}\left(1.01\times10^5\ \frac{N}{m^2}\right)(10^{-3}\ m^3) = \boxed{83.83\ J}$

g. Determine the net heat flow into / out of the gas during this process. Does the direction of heat flow make sense? (Compare the given process to adiabatic and isothermal processes.)

$\Delta E_{int} = Q - W$

$-126.1\ J = Q - 83.83\ J \rightarrow Q = -42.3\ J$



for monotomic gas, for $\gamma = \frac{5}{3}$ in $PV^\gamma = c$, for c a constant, $Q = 0$.

since $P = \frac{a}{V^2} \rightarrow PV^2 = a$ is a steeper (below) curve from $PV^{5/3} = c$, $Q < 0$.